

ERC Advanced Grant 2025
Research proposal [Part B1]¹

Cognitive Biases in Reasoning and Interpretation

NIHILPLUS

Cover Page:

- Name of the Principal Investigator (PI): Maria Aloni
- Name of the PI's Host Institution for the project: University of Amsterdam
- Proposal duration in months: 60 months

Proposal summary: (half page, max 2000 characters)

People often reason in ways which deviate from classical logic. An influential idea introduced by Grice is that these deviations are not logical mistakes but rather consequences of pragmatic enrichments derived as the product of rational interactions between cooperative language users. Challenging the Gricean tradition, the core hypothesis behind this project is that many of the enriched interpretations we observe in everyday conversation are not derived by Gricean reasoning, but rather result from biases due to our [human] preference to minimise cognitive effort. The main goal of the project is to arrive at a unified theory of such bias-driven interpretations and a principled explanation of how they relate to canonical pragmatic and semantic inferences in the context of a comprehensive account of human reasoning. We will study three biases that on my hypothesis affect both reasoning and interpretation: (i) a tendency to avoid emptiness (*neglect-zero*); (ii) a negative bias towards the parallel processing of multiple alternatives (*no-split*) and (iii) a tendency to assume that what is not known to be true is false (*min-mod*). Building on previous work of the PI the project will define **logics** that isolate the effect of these biases in deductions and interpretation; provide linguistic analyses detailing their impact in **natural language**, and conduct experiments probing their role in **cognition**. By combining logical modelling and experimental methods I aim at a novel architecture, where conventional, contextual and cognitive aspects of human reasoning and communication can be studied in a unified and principled way. In linguistics, cognitive biases have been argued to play a role but mostly to shape lexicalisations. In psychology and economics, the focus is often on their effect on fast thinking, (non-linguistic) decisions and behaviour. This would be the first project using logic to study their impact on enriched interpretations in both reasoning and conversation.

Cross-panel or Cross-domain Justification: (max 1000 characters)

This proposal lies at the interface of linguistics and philosophy. It addresses phenomena in the semantics and pragmatics of natural language (SH4 Human mind and its complexity), but crucially uses tools from (philosophical) logic (so it also relates to SH5_07 Philosophy of science, epistemology, logic).

¹Instructions for completing Part B1 can be found in the 'Information for Applicants to the Advanced Grant 2025 Call'.

Section a: Extended Synopsis of the scientific proposal (max. 5 pages)

State of the art, main hypotheses and objectives

If I tell my sons “You may do your homework or go to the beach” they conclude that they may go to the beach, contrary to the prescriptions of classical logic (classically, **may** $[\alpha \text{ or } \beta]$ does not entail **may** α). If asked to give examples of contradictions, my students often propose “If A then not A”, again deviating from classical rules (classically, the sentence is not a contradiction because it can be true, if A is false). What explains these common departures from classical reasoning? An attractive idea originating in the work of the philosopher Paul Grice is that these deviations are not logical mistakes, but rather a consequence of pragmatic enrichment: the fact that language users often interpret linguistic expressions as conveying more than what is literally said [Gri75]. In the Gricean tradition, pragmatic enrichments (also known as implicatures) are derived as the product of rational interactions between cooperative language users [Gaz79, Hor84, SW95, Lev00, FG12]. More recently, the “grammatical” view challenged the Gricean approach and proposed that some implicatures follow from optional applications of covert grammatical operators [Chi06, Fox07, CFS12, BLF20]. On both views, it’s rational requirements that drive pragmatic enrichment. Principled rules select the “most optimal” interpretation among a set of alternatives: a step of rational choice. Departing from both these traditions, the **novel hypothesis** at the core of this proposal is that, in many cases, it’s not rational demands that govern pragmatic enrichment but rather rational “defects”: many of the enriched interpretations we observe in everyday conversation are not the product of rational choices, but result from *biases* due to our [human] cognitive limitations. The project sets to study two specific biases, due to our preference to minimize cognitive effort, which I propose affect both deductive reasoning and linguistic interpretation: (i) a tendency to avoid emptiness (*neglect-zero*); and (ii) a negative bias towards the parallel processing of multiple alternatives (*no-split*). I will further explore their interplay with other biases, notably, our preference for minimal models (*min-mod*), which has been argued to explain non-monotonic reasoning [SvL08] and phenomena of exhaustification [GS84]. Combining LOGICAL and EXPERIMENTAL methods, the **main objective** of the project is to arrive at a unified theory of bias-driven interpretations and hereby a principled explanation of how they relate to canonical pragmatic and semantic inferences in the context of a comprehensive account of everyday reasoning and its deviations from its logical/mathematical counterpart. By paying due attention to the cognitive aspects impacting human communication the project will substantially advance research in formal pragmatics where these aspects are usually neglected. By defining logics that model cognitive biases and derive their effect in deductions and interpretation it will also contribute to existing research in the psychology of reasoning whose descriptions of biases and their impact often lack rigorous formalisation.

Neglect-zero The first bias we set to study is the neglect-zero tendency first presented by the PI in [Alo22].

NEGLECT-ZERO HYPOTHESIS

When interpreting a sentence speakers construct models depicting reality and in this process tend to neglect models that verify the sentence by virtue of an empty configuration (*zero-models*).

The neglect-zero tendency, which connects to a general preference in human cognition for concrete (non-empty) over abstract (empty) representations [Pai65], follows from the expected difficulty of the cognitive operation of evaluating truths with respect to empty witness sets [BSK19]. Structures that verify a sentence by virtue of an empty set are called *zero-models*. As an illustration consider the following examples:

- | | |
|--|--|
| <p>(1) Every square is black.</p> <p>a. Verifier: $[\blacksquare, \blacksquare, \blacksquare]$</p> <p>b. Falsifier: $[\blacksquare, \square, \blacksquare]$</p> <p>c. Zero-models: $[\triangle, \triangle, \triangle]; [\blacktriangle, \blacktriangle, \blacktriangle]; \dots$</p> | <p>(2) Less than three squares are black.</p> <p>a. Verifier: $[\blacksquare, \square, \blacksquare]$</p> <p>b. Falsifier: $[\blacksquare, \blacksquare, \blacksquare]$</p> <p>c. Zero-models: $[\square, \square, \square]; [\blacktriangle, \blacktriangle, \blacktriangle]; \dots$</p> |
|--|--|

One highly influential tradition in psychology and psycholinguistics, i.e. the mental model tradition [JL83], builds on the assumption that the interpretation of (1) and (2) leads to the creation of internal structures representing reality, some verifying the sentence (the models in (a)), and some falsifying it (those in (b)). The novel neglect-zero hypothesis states that zero-models, the ones represented in (c), are usually kept out of consideration. The models in (1-c) and (2-c) are zero-models for (1) and (2), respectively, because they verify the sentence by virtue of an empty set, an empty set of squares in the case of (1-c) and an empty set of black squares for (2-c). In the model-theoretic tradition in logical semantics, deductive reasoning involves checking whether the conclusion of an argument is satisfied in all verifiers of the premises. The exclusion of zero-models from the set of verifiers leads directly to an account of natural language inferences such as the existential import (EI) effects operative in the logic of Aristotle [e.g., *every square is black* $\rightsquigarrow$ *some square is black*, AR02, Geu07, Man23].

Zero-models are neglected because they are cognitively taxing. Findings from number cognition confirm this difficulty [RCMN16], which also explains the special status of the zero among the natural numbers [e.g.,

its late invention in human history, late emergence in human development, and special representation in the brain, Nie16, KKL21, BF24] and why downward-monotonic quantifiers (e.g., *less than n squares*) are more difficult to process than upward-monotonic ones (e.g., *more than n squares*) [BSK19]. Since empty witnesses encode the absence of objects/properties, they are more detached from sensory experience and therefore harder to conceive. The inference from the perception of absence to the truth of a sentence brings in additional costs, which results in a systematic negative bias towards zero-models, a neglect-zero tendency.

The first idea at the core of this proposal is that several enriched interpretations, prominently explained by neo-Gricean or grammatical mechanisms, are instead, just like the Aristotelian EI effects, a consequence of such a neglect-zero tendency assumed to be operative in ordinary conversation. Recent work of the PI [Alo22, Alo23, AvO23] detailed how neglect-zero can derive classically non-valid inferences arising from disjunctive and quantified statements, such as Free Choice (FC) inferences where disjunctions lead to conjunctive meanings contrary to the rules of classical logic [Kam73, Zim00, Kli06, Alo07b, Fox07, Fra11, Gol19, CAG19, a.o.]:

(3) **Disjunction**

- a. You may go to the beach *or* to the cinema $\rightsquigarrow$ you may go to the beach *and* you may go to the cinema [FC]
- b. Every square is red *or* white $\rightsquigarrow$ there are white squares *and* there are red squares [Distribution (DIST)]
- c. The prize is either in the garden *or* in the attic $\rightsquigarrow$ speaker doesn't know where [Ignorance (IGN), Gri89]

(4) **Empty Set (ES) Quantifiers**

- a. Every square is black $\rightsquigarrow$ there are some squares [ESQ-restrictor]
- b. Less than 3 squares are black $\rightsquigarrow$ there are some black squares [ESQ-scope]

On the Gricean and grammatical view all these inferences are, like scalar implicatures (e.g., the inference from *some* to *not all*), typically derived through a pragmatic or grammatical operation that negates potential alternatives. My hypothesis instead is that FC, DIST, IGN and ESQ-inferences are essentially different from scalar implicatures and a direct consequence of a neglect-zero bias.

	Ignorance	FC & DIST	Scalar implicature	ES-Quantifiers
Neo-Gricean	reasoning	reasoning	reasoning	reasoning
Grammatical view	debated	grammatical	grammatical	grammatical
NIHILPLUS	neglect-zero	neglect-zero	—	neglect-zero

Recent EXPERIMENTS provided initial support for this hypothesis. First, scalar implicatures are acquired late [Nov01, GCM⁺01, PM03], and take time to process [BN04]. This is not the case for FC inferences, which are acquired early [TRZC16], and their processing time equals that of literal interpretations [CB14]. Second, recent collaborative work of the PI [KSB⁺25] found evidence of semantic priming between DIST and ESQ-scope inferences but no priming between scalar and ESQ-scope inferences indirectly confirming previous results by [MF21] who found across-construction priming between two types of scalar inferences but no priming between FC and scalar *some* inferences. By providing evidence of a common mechanism in use for deriving DIST and ESQ-scope inferences (but not scalar implicatures) these results support the hypothesis of a divide between Disjunction and ES-Quantifiers inferences on one side and scalar inferences on the other in agreement with the neglect-zero hypothesis. Third, another collaborative work of the PI [DMR⁺25] studied ignorance inferences with findings challenging the standard Gricean account but in agreement with the neglect-zero hypothesis.

The present project will substantially expand this initial research on neglect-zero by investigating more constructions impacted by this bias (e.g., conditionals, see (5)), but it will also go beyond this work, while building on its foundations, and break new ground by introducing a novel cognitive bias, **no-split**, and by investigating interplays with other biases (notably **min-mod**, the tendency to prefer minimal models discussed by [SvL08]) with the ultimate goal to arrive at a comprehensive theory of bias-driven interpretation and a systematic comparison of the cognitive bias view with the alternative neo-Gricean and grammatical approaches.

(5) **Conditionals**

- a. If it is raining, it is not raining $\rightsquigarrow \perp$ (contradiction) [cf. Aristotle's Thesis, Pri99]
- b. If it is raining, you should bring an umbrella $\rightsquigarrow$ it might be raining

No-split To illustrate the impact of neglect-zero on our interpretation of disjunction consider first the positive episodic sentences (6) and (7). On my hypothesis their interpretation involves the construction of mental models verifying the sentence (in (a)) and falsifying it (in (b)). Notice that there are no zero-models for such positive sentences. Consider now the disjunction in (8):

- | | | |
|--|--|--|
| <p>(6) It is raining.</p> <ul style="list-style-type: none"> a. Verifier: [//// //] b. Falsifier: [☀☀☀☀] c. Zero-models: none | <p>(7) It is snowing.</p> <ul style="list-style-type: none"> a. Verifier: [***] b. Falsifier: [☀☀☀☀] c. Zero-models: none | <p>(8) It is raining or snowing.</p> <ul style="list-style-type: none"> a. Verifier: [//// // ***] $\Leftarrow$ split b. Falsifier: [☀☀☀☀] c. Zero-models: [//// //]; [***] |
|--|--|--|

The sentence has two *zero-models* represented in (8-c). These are zero-models because they verify the sentence by virtue of an empty witness for one of the disjuncts. For example, the model $[\text{rain} \text{ } \text{snow}]$, depicting rain, relies on an empty witness for the second disjunct that evokes the possibility of snow. On my hypothesis, such zero-models are cognitively taxing and therefore disregarded as potential verifiers for (8). What speakers instead construct when interpreting (8) are models like the one in (8-a), where different (possibly conflicting) alternatives are simultaneously entertained. (8-a) is a model where both rain and snow are open possibilities [Zim00, Geu05]. I call these models *split-states*. On my neglect-zero hypothesis, IGN and FC effects arise because split-states emerge as natural verifiers for disjunctions since zero-models, where only one of the disjuncts is depicted, are kept out of consideration. Split-states, where multiple alternatives are processed in a parallel fashion, however, are also cognitively taxing because of the increased effort they impose on working memory (WM). My second hypothesis is that, given their increased WM load, split-states are (i) particularly challenging for children who systematically avoid them [LHS⁺22]; and (ii) they are also difficult for adults who tend to disprefer them but only in environments which bring in additional complexity, e.g., under negation [Sba25].

NO-SPLIT CONJECTURE

- (i) the ability to split states (entertain multiple alternatives) is developed late;
- (ii) also adults tend to avoid split-states in cognitively taxing environments, notably under negation.

Recent collaborative work of the PI [KSA25] showed that the combination of neglect-zero and no-split can explain (i) non-classical inferences observed in pre-school children such as (9), where a disjunction is interpreted as a conjunction [SWAR⁺16, CDvH23, CDvH24, BTB⁺25, and older data reported by [EBN93]]; but also (ii) homogenous (HOM) interpretations of conjunctions under negation as illustrated in (10) [Kri15, BL18, Sba25]:

- (9) The boy is holding an apple *or* a banana $\rightsquigarrow$ the boy is holding an apple *and* a banana [Conjunctive *or*]
- a. Split & no-zero verifier: [ | ] (adults)
 - b. No-split & no-zero verifier: [ & ] (children)
- (10) The boy is *not* holding an apple *and* a banana $\rightsquigarrow$ the boy is *neither* holding an apple *nor* a banana [HOM]
- a. Split & no-zero verifier: [NO  | NO ] (adults, classical interpretation)
 - b. No-split & no-zero verifier: [NO  & NO ] (adults, homogeneous interpretation)

For children, given their limited WM capacity, the combined cost of zero-models and split-states leads to a preference for (9-b), the only possible no-zero and no-split model for disjunction (9). This is a model where both disjuncts are true yielding a conjunctive interpretation. Adults, who have higher WM capacity (and possibly a better knowledge of the $\langle \text{or}, \text{and} \rangle$ lexical scale [BBB11]), can better balance between the two opposing forces of cognitive simplicity and informative communication [KXR18] and select split-states for (9), but no-zero and no-split states reemerge as the preferred option in (10) because, we conjecture, of the additional effort brought in by negation [KOJL12] yielding homogeneity effects in adults' interpretations.

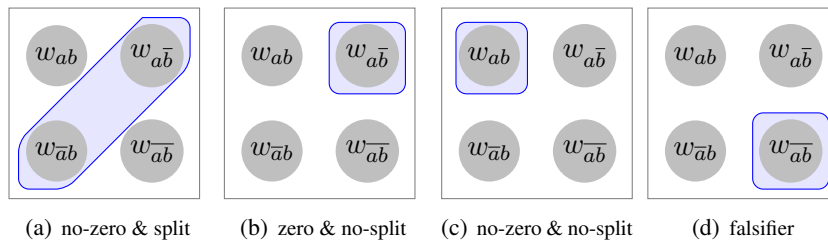


Figure 1: Models for $(a \vee b)$.

Logical modelling As shown in previous work of the PI [Alo22, AAY24], the biases informally presented above can be rigorously formalised in LOGICS using team semantics [Vö7, YV17]. In *team semantics*, logical formulas are interpreted with respect to *sets* of points of evaluation (teams) rather than single points. Two features of this framework are crucial for our purposes: (i) the empty set is one of the possible teams and can be employed to model zero-models; (ii) multi-membered teams can model the simultaneous engagement with contrasting alternatives (split states). Given (i), **neglect-zero** can be modelled in team semantics by disallowing empty teams as possible verifiers. Given (ii), **no-split** can also be modelled, e.g., via ‘flattening’ operators which induce pointwise evaluations and therefore prevent simultaneous processing of alternatives. By introducing neglect-zero and no-split *enrichment functions*, we can arrive at precise definitions of the notion of (no) zero-models and (no) split-states for any given sentence (see [Alo22] for zero-models and [KSA25] for a first extension to no-split). Figure 1 gives an illustration of teams (the blue areas) representing different kinds of verifiers or falsifier for the disjunction $a \text{ or } b$ ($a \vee b$) in our formalisation. In Fig. 1, w_{ab} is a world where a and b are both true, $w_{a\bar{b}}$ one where a is true and b is false, ect. Note that all no-zero verifiers are teams where both

a and b are open possibilities (which derives IGN and FC inferences); and the no-zero & no-split verifier in (c) additionally supports the inference that both a and b are true (deriving children’s conjunctive interpretations).

Interplay with other biases We assume that both reasoning and understanding of natural language involve the creation of mental structures. Understanding a sentence means being able to mentally construct a model picturing the world which verifies the sentence, and possibly also a model which falsifies it. Deductive reasoning depends on two main processes: first we construct verifying models for the premises and then the validity of the conclusion is checked on these models. What the current project sets to investigate are biases which constrain the construction of these models and therefore impact both reasoning and interpretations: the **neglect-zero** bias prevents the constructions of zero-models. The **no-split** bias expresses a dispreference for split-states. [SvL08] argued that in deductive reasoning subjects usually do not consider all models of the premises, but only *minimal* ones, i.e., models verifying the premises and falsifying anything else. The tendency to assume that what is not known to be true is false (i.e., a preference for minimal models, **min-mod**) has the potential to derive non-monotonic reasoning [SvL08], exhaustifications [GS84] but also maybe scalar implicatures [vRS04, Alo07a]. With the aim to explore the interplay of min-mod with our 2 biases and its potential to capture scalar reasoning, I will develop a **new logical framework**, combining tools from team semantics and logic programming, where all 3 biases can be modelled. The ultimate goal is the development of a comprehensive theory where different types of non-classical inferences (Free Choice, Homogeneity, Scalar, ...) find their natural place.

	Disjunction & ES-Quantifiers	Conditionals	Scalar implicature	Conjunctive <i>or</i>	Homogeneity
Neo-Gricean	reasoning	reasoning	reasoning	—	—
Grammatical view	grammatical	—	grammatical	grammatical	grammatical
NIHILPLUS	neglect-zero	neglect-zero	min-mod [?]	negl-z + no-split	negl-z + no-split

Table 1: Summary main hypotheses and comparison with alternative views

Objectives and Methods The **overall objective (OO)** of this research is to arrive at a general theory of bias-driven interpretations and hereby a principled explanation of how they relate to canonical pragmatic and semantic inferences in the context of a comprehensive account of human reasoning. The **general strategy** is *interdisciplinary* in nature: (i) define logics which model next to literal meanings (ruled by classical logic), also cognitive (and contextual) factors, and the additional inferences derived by their interplay; (ii) feed these logics with detailed linguistic analyses of the phenomena in Table 1; and (iii) conduct experiments to test the predictions arising from these analyses and rigorous formalisations. Five **subgoals** contribute to achieving **OO**: [G1] Formulate novel hypotheses delineating the impact of neglect-zero, no-split, min-mod and their interplay; [G2] Detail the predictions arising from these hypotheses via linguistic analyses and logical modelling; [G3] Test these predictions with experimental studies; [G4] Compare and integrate the cognitive bias view with alternative approaches; [G5] Explore interfaces with neighbouring disciplines.

Innovation, Groundbreaking Nature, and Impact The project breaks new ground by proposing a disruptive hypothesis that challenges the neo-Gricean and grammatical perspectives which currently dominate research in formal pragmatics. An equally innovative aspect of the project is the use of logics to model cognitive biases and isolate their impact on communicative inferences. By defining predictive theories of the role of biases in reasoning and conversation the project substantially advances research in formal pragmatics (which traditionally neglects cognitive aspects) and in the psychology of reasoning (which addresses cognitive concerns but often lack rigorous formalisations). In linguistics, cognitive biases have already been argued to play a role but mostly to shape lexicalisations [KR12, Lev12, KXR18, CSTS21, ES21]. In psychology and economics, the focus is often on belief-based or matching biases and their effects on fast thinking, (non-linguistic) predictions, decisions and behaviour [TK74, And91, Eva03, Kah11, DN12]. In none of these traditions, cognitive biases received a logical characterisation. This would be the first project to use logic to investigate the impact of cognitive biases on ordinary language use. If successful it has the potential to reshape our understanding of how semantic, pragmatic and cognitive aspects are *integrated* in everyday reasoning and communication.

Structure of the project, research team, and overview of the work packages

The main objectives identified above will be pursued in the following **work packages (WPs)**.

WP1-Foundations [PI; PD1] $\mapsto$ G1, G5. The main goal of WP1 is to provide theoretical and logical underpinning for the research carried out in other WPs. By careful analytical work and logical modelling WP1 will produce novel hypotheses, which WP2 will further develop to generate precise predictions which will then be tested experimentally in WP3. WP1 will specifically address the question of how our 3 biases interact, and explore interfaces with neighbouring disciplines. To this purpose it will define (a) multilateral proof-theories [IS22, AIS23] for team-based logics enhancing their potential for applications in the *psychology of reasoning*

[e.g., KBJL18, KM13]; and (b) algebraic characterisations of these logics [BBH16, vB19], providing a framework for rigorous comparison with related systems in *philosophical logic* (truthmaker, inquisitive & possibility semantics, relevance logic [Fin17, CGR18, Hum81, AB75]). *Methods*: Conceptual analysis, logical methods.

WP2-Logic & Language [PHD1; PHD2] $\mapsto$ G2, G4. The main goal of WP2 is to provide detailed linguistic analyses of the phenomena in Table 1 but also rigorous logical modelling (mostly model-theoretical) and in depths comparisons with alternative views which have been proposed in formal pragmatics focusing mostly on the most recent representatives of the Gricean approach [Rational Speech Act (RSA), GS12, CAG19, CWS23] and of the grammatical view [SWAR⁺16, BLF20, BDPS21]. PHD1 will focus on neglect-zero & min-mod; PHD2 on no-split. In the last phase PHD1 will work on the possible integration within the cognitive bias framework of RSA and grammatical mechanisms to account for gradience, particularized conversational implicatures [Gri75] and the embeddings of scalar inferences [CFS12]. *Methods*: Linguistic analysis, logical modelling.

WP3-Cognition [PD2] $\mapsto$ G3. The main goal of WP3 is to experimentally test the hypotheses of the project. It will conduct (i) online *priming* studies [PF08, FS16, BC16, MRSB24, KSB⁺25] to uncover common or diverging cognitive mechanisms behind the different inferences in Table 1; (ii) experiments of different kinds (*dual task* (DT) [VTPS19, RA25], *eye-tracking* (gaze fixation), *pupillometry* [RSS⁺25, MSKH24]) to measure the hypothesised cognitive difficulty of zero- and split-models which is at the core of my cognitive biases proposal; and (iii) longitudinal *development/acquisition* experiments to (a) explore the connections between children (4-6 years old)'s knowledge of the number zero [KKL21, MB13] and their engagement with zero-models in linguistic interpretation [GKC⁺10]; and (b) test the predictions of the no-split hypothesis on the acquisition of *or* [KSA25]. At the end of WP3, we leave time for experimentally testing further specific diverging predictions arising from the comparative linguistic studies carried out in WP2 (notably on the impact of min-mod). All experiments will be conducted by PD2 in collaboration with experts with whom I already collaborated. These include J. Romoli (Düsseldorf, priming), O. Bott (Bielefeld, priming, eye-tracking), F. Schlotterbeck (Tübingen, priming, eye-tracking), J. Schaeffner (ILLC, acquisition), I. Visser (UvA, development).

The 3 WPs will not just be carried out side by side, but will interact systematically. On the one hand, the foundational investigation in WP1 will produce hypotheses that will be detailed by the logics developed in WP2 which will produce predictions that will be tested in the experimental WP3. On the other hand, the findings of WP3 will inspire new theorising, which will be constrained and made focused by logical concerns. **WP4-Synthesis** [PI] $\mapsto$ OO will synthesise the results of the WPs in a unified theory of bias-driven interpretation.

Feasibility & Timeline The proposed interdisciplinary approach combining logical and experimental methods has been proven successful in my recent research leading to substantial advances in logic, linguistics and cognitive science [AAY24, DMR⁺25, KSB⁺25]. Therefore, despite covering uncharted territory, the planned research is feasible and even in the unexpected event that all my hypotheses will be falsified, it will substantially advance our understanding of the impact of cognitive aspects in the derivation of communicative inferences.

NIHIL ⁺	PI (Aloni) [WP1&WP4]	PD1 [WP1]	PhD1 [WP2]	PhD2 [WP2]	PD2 [WP3]
Y1	1 Preparatory work				
	2 Close supervision & team building	Multilater. proof-theory [negl.-zer] (prop, modals & quantifiers)	Initial training	Initial training	Prepar. work
	3 & team building				
	4 Kick-off workshop		Negl.-zero (adding conditionals)	No-split (modeling)	Negl.-zero exp (prim., DT, eye-track, acquisit.)
Y2	1 Neglect-zero & no-split (foundations & integration)	Multilateral Proof Theory [no-split] (child; adult)	Conditionals, Or & Quantifiers (analysis & comp.)	Acquisition of disjunct. (analysis & comparisons)	No-split exp (acquisition, DT, eye-track, priming)
	2 Minimal Models (modeling & integration)				
	3				
	4				
Y3	1 Mid-term workshop	Algebraic character. & comparisons via translations	Scalar Implicatures (min-mod, RSA & grammatical view)	Homogeneity (analysis & comparisons)	
	2 Psychology of reasoning research (reflection & integration)		Partic. conv. impl. & gradience (RSA)		Acquisition & further (min-mod) experiments
	3				
	4				
Y4	1 Unified Theory Bias-driven Interpretation (final synthesis)				
	2				
	3				
	4 Final workshop		Dissertation writing	Dissertation writing	
Y5	1				
	2 Editing volume: Synthesis, reflection & repercussions				
	3				
	4				

Conclusion The ultimate goal of the project is to place common departures from classical reasoning on rigorous **foundations** by defining **logics** that isolate the effect of cognitive biases in deductions and interpretations. This logical enterprise is supplemented by linguistic analyses studying the impact of these biases in **natural language**, and by experiments probing their role in **cognition**. By combining foundational research, logical modelling, and linguistic and experimental studies I aim at a novel architecture where conventional, contextual and cognitive aspects of human reasoning and communication can be studied in a rigorous and principled way.

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